Impressionist Painting Generation Using WGAN-GP

Zichen Hu (zih050@ucsd.edu)
University of California, San Diego

1 Hyperparameter Settings and Model Training

We implement a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) in PyTorch to generate Impressionist paintings. The dataset is taken from WikiArt. To reduce computational cost, we select the Impressionism category, containing 10,121 training images and 2,536 testing images. All images are resized to 64×64, converted to RGB, and normalized to $[-1, 1]$.

The generator uses five transposed convolutional layers to transform a 128-dimensional Gaussian latent vector into a painting. The critic uses convolutional layers with LeakyReLU activations and layer normalization to distinguish real and generated image distributions.

After exploring different GAN configurations, WGAN-GP was selected for the final experiment based on the visual appearance of its generated samples. Its gradient penalty helps enforce the Lipschitz constraint and stabilize adversarial training.

The model is trained for 60 epochs using Adam with a learning rate of 0.0001, batch size of 64, and gradient penalty coefficient of 10. The critic is updated three times per generator update.

2 Training Loss Curves

Figure 1 shows the generator loss, critic loss, and gradient penalty throughout training.

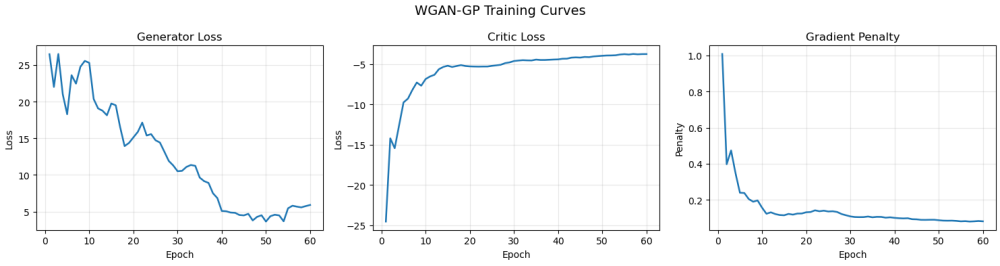


Figure 1: WGAN-GP training losses over 60 epochs.

The generator loss decreases substantially during training, while the critic loss gradually stabilizes. The gradient penalty also decreases and remains relatively stable in later epochs.

At epoch 60, the generator loss is 5.9025, the critic loss is -3.7705, and the gradient penalty is 0.0806. These trends indicate stable numerical training, although the loss values alone do not measure image quality.

3 Inception Score and FID Evaluation

We evaluate generated images using Fréchet Inception Distance (FID) and Inception Score (IS). A total of 2,536 generated paintings are compared against 2,536 real test images.

Both metrics use an ImageNet-pretrained InceptionV3 network. FID is calculated using 2,048-dimensional pooled features, while IS uses predicted class probabilities.

We evaluate checkpoints from epochs 40, 50, and 60. The epoch-60 checkpoint achieves the lowest FID among the evaluated WGAN-GP checkpoints and is selected for final image generation.

FID: 127.1088 Inception Score: 2.2022

The results indicate that the model learns meaningful painting characteristics, although the distribution of generated images still differs from real WikiArt paintings. These metrics are useful for evaluation but do not fully represent artistic quality.

4 Generated Painting Examples

Figure 2 shows 16 paintings generated by sampling latent vectors from $z \sim \mathcal{N}(0, I)$ and passing them through the trained generator.

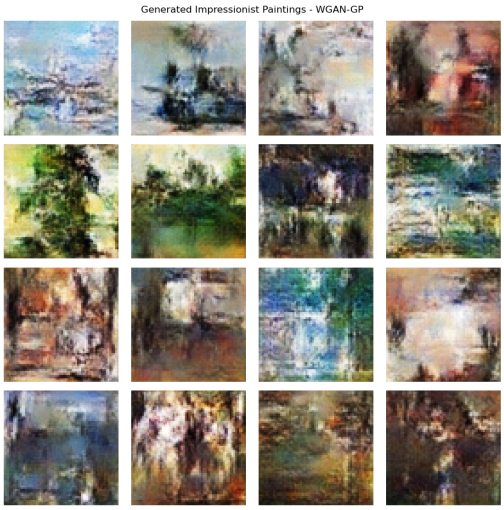


Figure 2: Impressionist paintings generated by WGAN-GP.

The generated images contain diverse colors, brushstroke-like textures, and landscape structures. Some resemble trees, water, and natural scenery. However, fine details remain blurry, and certain objects are difficult to recognize.

Overall, WGAN-GP successfully generates varied painting-like images, although further improvements would be needed to produce more realistic and structurally coherent artwork.

Code and Results: `https://github.com/witnesszc23/ECE-285`